

Solving and Classifying Linear Equations

One solution, no solution, or infinitely many solutions

Directions

- Simplify each side of the equation first, then collect the variable terms on one side.
- The step where the variable terms combine or cancel is what tells you the classification — show it.
- Write each final answer as a value of the variable, or as *no solution*, or as *infinitely many solutions*.

1. Bike Barn rents bicycles for a \$9 flat fee plus \$4 per hour. Wheel World charges a \$14 flat fee plus \$3 per hour. The equation $9 + 4h = 14 + 3h$ compares the two rental costs in dollars, where h is the number of hours. Solve it, then state how many solutions it has.

Answer: _____

2. One phone plan costs \$12 per month plus \$6 for each gigabyte of data used. A second plan costs \$20 per month plus \$4 for each gigabyte. Solve $12 + 6g = 20 + 4g$, where g is the number of gigabytes used in a month, then state how many solutions it has.

Answer: _____

3. A gardener has r rows of 5 plants and then adds 3 more rows of 5 plants. Counting by groups gives $5(r+3)$ plants; counting row by row gives $5r+15$ plants. Solve $5(r+3) = 5r+15$, then state how many solutions it has.

Answer: _____

4. Savings Plan A starts with \$25 and adds \$5 every week. Savings Plan B starts with \$18 and adds \$5 every week. The equation $25 + 5w = 18 + 5w$ asks when the two balances in dollars are equal, where w is the number of weeks. Solve it, then state how many solutions it has.

Answer: _____

5. A club spends $8(t - 2)$ dollars on snacks and $6t + 4$ dollars on supplies, where t is the number of members. Solve $8(t - 2) = 6t + 4$ to find the membership at which the two amounts are equal, then state how many solutions the equation has.

Answer: _____

6. A rectangle is x cm long and 4 cm wide. Mia writes its perimeter as $2(x + 4)$ cm; Ben writes it as $2x + 8$ cm. Solve $2(x + 4) = 2x + 8$, then state how many solutions it has.

Answer: _____

7. On a track, runner A starts 30 m from the line and runs 6 m/s, so her distance is $30 + 6t$ metres after t seconds. Runner B starts 12 m from the line and also runs 6 m/s: $12 + 6t$ metres. Solve $30 + 6t = 12 + 6t$, then state how many solutions it has.

Answer: _____

8. A pool is draining and holds $120 - 8m$ litres after m minutes. A second pool is filling and holds $40 + 12m$ litres after m minutes. Solve $120 - 8m = 40 + 12m$ to find when the two pools hold the same amount, then state how many solutions the equation has.

Answer: _____

9. A print shop charges $4(2x + 5)$ dollars for x posters. A rival shop charges $8x + k$ dollars, where k is a whole number of dollars.

- (a) Find the value of k that makes $4(2x + 5) = 8x + k$ true for every number of posters.

Answer: _____

- (b) Explain how the simplified equation shows that every other value of k gives an equation with no solution.

Answer: _____

10. Tank A holds $60 + 9t$ litres of water after t minutes. Tank B holds $3(3t + 12)$ litres after t minutes.

- (a) Simplify both sides of $60 + 9t = 3(3t + 12)$ and state how many solutions the equation has.

Answer: _____

- (b) Explain what the simplified equation tells you about the two water amounts as the minutes go by.

Answer: _____